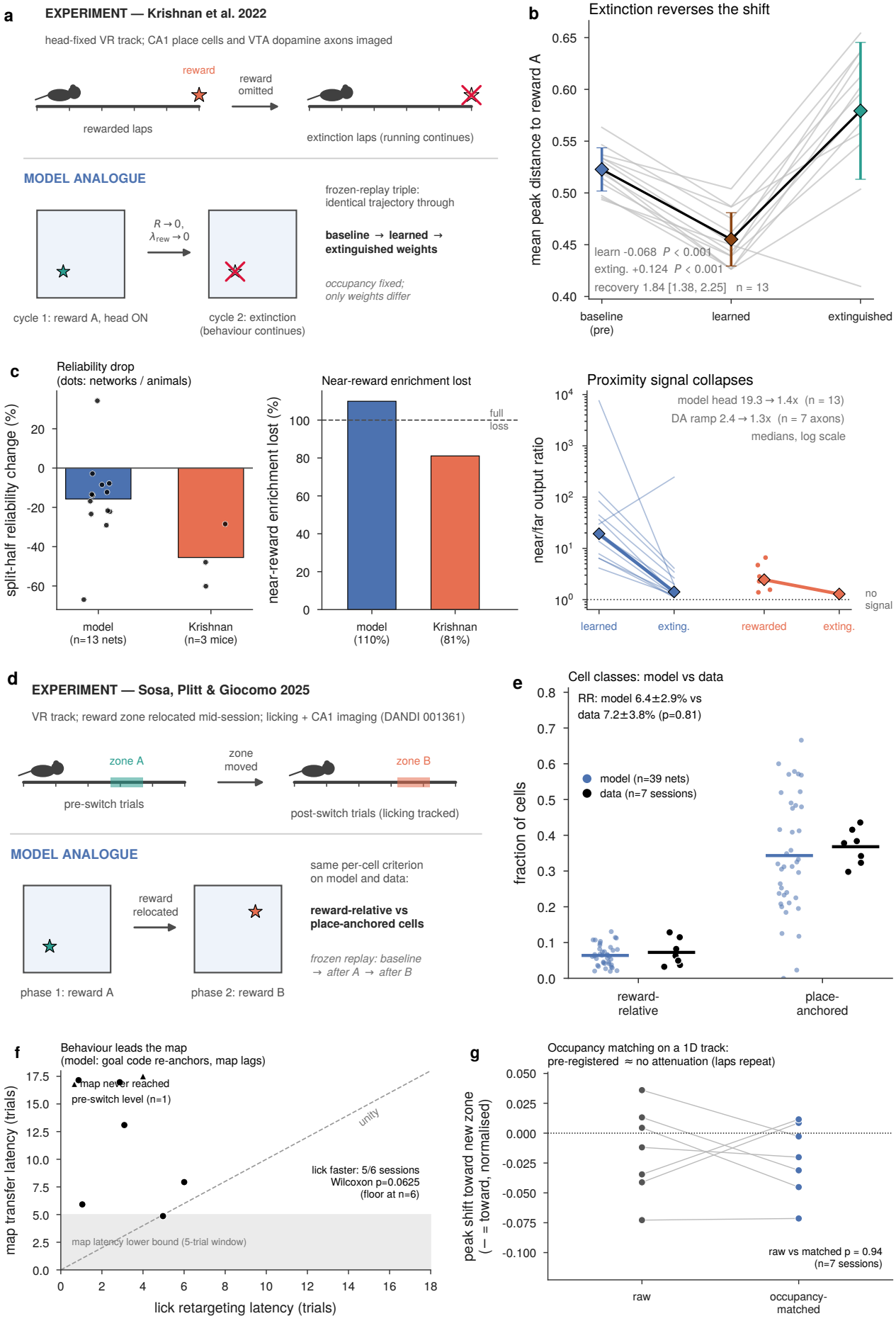

a
EXPERIMENT — Krishnan et al. 2022
head-fixed VR track; CA1 place cells and VTA dopamine axons imaged
reward
reward omitted
rewarded laps
extinction laps (running continues)
MODEL ANALOGUE
R → 0, λrew → 0
cycle 1: reward A, head ON
cycle 2: extinction (behaviour continues)
frozen-replay triple: identical trajectory through
baseline → learned → extinguished weights
occupancy fixed; only weights differ
b
Extinction reverses the shift
mean peak distance to reward A
learn -0.068 P < 0.001
exting. +0.124 P < 0.001
recovery 1.84 [1.38, 2.25] n = 13
baseline (pre)
learned
extinguished
c
Reliability drop
(dots: networks / animals)
split-half reliability change (%)
model (n=13 nets)
Krishnan (n=3 mice)
Near-reward enrichment lost
near-reward enrichment lost (%)
full loss
model (110%)
Krishnan (81%)
Proximity signal collapses
model head 19.3 → 1.4x (n = 13)
DA ramp 2.4 → 1.3x (n = 7 axons)
medians, log scale
near/far output ratio
no signal
learned
exting.
rewarded
exting.
d
EXPERIMENT — Sosa, Plitt & Giocomo 2025
VR track; reward zone relocated mid-session; licking + CA1 imaging (DANDI 001361)
zone A
zone moved
zone B
pre-switch trials
post-switch trials (licking tracked)
MODEL ANALOGUE
reward relocated
phase 1: reward A
phase 2: reward B
same per-cell criterion on model and data:
reward-relative vs place-anchored cells
frozen replay: baseline → after A → after B
e
Cell classes: model vs data
RR: model 6.4±2.9% vs data 7.2±3.8% (p=0.81)
model (n=39 nets)
data (n=7 sessions)
fraction of cells
reward-relative
place-anchored
f
Behaviour leads the map
(model: goal code re-anchors, map lags)
map never reached pre-switch level (n=1)
unity
lick faster: 5/6 sessions
Wilcoxon p=0.0625
(floor at n=6)
map latency lower bound (5-trial window)
map transfer latency (trials)
lick retargeting latency (trials)
g
Occupancy matching on a 1D track:
pre-registered ≈ no attenuation (laps repeat)
peak shift toward new zone (− = toward, normalised)
raw vs matched p = 0.94
(n=7 sessions)
raw
occupancy-matched